Prime-Support Dilution in Consecutive Integer Products

Conjectures, finite tests, unconditional sparsity, and possible routes to a uniform proof

Research working note developed in human–AI mathematical discussion

22 October 2026
Version 3: Result 012 review
Private research working draft

Abstract

For $N = m(m+1)$, let $\omega(N)$ be its number of distinct prime factors and let $\iota(N) = \pi(P^+(N))$ be the index of its greatest prime factor. We propose *prime-support dilution*: $\omega(m(m+1))/\iota(m(m+1)) \to 0$ as $m \to \infty$, and the more speculative bound $\omega(m(m+1)) \le \iota(m(m+1))^{1/2+\varepsilon}$ for every $\varepsilon > 0$ and sufficiently large m. Both statements are stronger than finiteness of prime-complete consecutive products (OEIS A141399). We give a new, self-contained derivation of an *unconditional* sparse-exception statement: for each fixed $\delta > 0$, the number of $m \le X$ for which at least a δ-fraction of the initial prime indices occur is $X^{o(1)}$. We also give an elementary unbounded lower bound for the maximal support function. Conditional on the strong Lagarias–Soundararajan *xyz* conjecture, an exponent-2/3 upper bound follows for the maximal number of prime divisors of consecutive p_r-smooth integers. A comparison with Erdős Problem 368 shows that its conjectural lower bound $P^+(m(m+1)) \gg (\log m)^2$ would imply our square-root variant. We examine OpenAI's September 2026 Result 012, which claims a natural-density joint Dickman theorem for consecutive integers, and carefully distinguish its fixed power-scale conclusions from the uniform logarithmic-scale estimates needed here. New exhaustive finite enumeration through $r = 20$ gives the exact values of that maximum and supplies a reproducibility check against two OEIS sequences. The small-order data do *not* yet provide convincing evidence for an asymptotic exponent 1/2. We separate rigorous results, conditional implications, heuristic arguments, and the unproved step needed for a proof independent of *xyz*.

1 Motivation and exact notation

Let $p_1 = 2 < p_2 = 3 < \cdots$ denote the primes, and set

$$P_r = p_r^{\#} = \prod_{i=1}^{r} p_i, \quad \omega(n) = \#\{p : p \mid n\}, \quad \iota(n) = \pi(P^+(n)), \quad n > 1.$$

Here p_r-smooth means *no prime factor greater than* p_r. In particular, *19-smooth is* p_8*-smooth*, not p_{19}-smooth. For $m \ge 1$ put

$$s(m) = \omega(m(m+1)), \qquad r(m) = \iota(m(m+1)), \qquad D(m) = \frac{s(m)}{r(m)}, \qquad \Delta(m) = r(m) - s(m).$$

The value $D(m)$ lies in $(0, 1]$. We call $D(m)$ the *prime-support fraction*. A product $m(m+1)$ is prime-complete (PC) precisely when

$$\mathrm{rad}(m(m+1)) = P_{r(m)} \quad \Longleftrightarrow \quad D(m) = 1.$$

The known values of m satisfying this equation form OEIS A141399 [21]; the largest known is 633555. Its global maximality is unproved. The present proposal changes the focus from the absence of *complete* support to the asymptotic *fraction* of support attained.

For each $r \geq 1$ define the finite Størmer set and its two extremal statistics:

$$\begin{aligned} \mathcal{S}_r &= \{m \geq 1 : P^+(m(m+1)) \leq p_r\}, \\ U_r &= \max \mathcal{S}_r = \mathrm{A002072}(r), \\ W(r) &= \max_{m \in \mathcal{S}_r} \omega(m(m+1)). \end{aligned}$$

Størmer's theorem guarantees that $\mathcal{S}_r$ is finite for fixed r [17, 18]. Note that $W(r)$ is *not* the maximum ω in a primorial interval. It maximizes over *all consecutive p_r-smooth pairs*, regardless of their size.

2 A hierarchy of conjectures

Conjecture 2.1 (Prime-support dilution)**.**

$$\boxed{\lim_{m \to \infty} D(m) = 0.}$$

Equivalently, the fraction of missing primes below the greatest prime factor satisfies $\Delta(m)/r(m) \to 1$.

Conjecture 2.2 (Power-saving dilution)**.** *There exists $\eta > 0$ such that, for all sufficiently large m,*

$$\omega(m(m+1)) \leq \iota(m(m+1))^{1-\eta}.$$

Conjecture 2.3 (Square-root dilution; speculative)**.** *For every $\varepsilon > 0$, for all sufficiently large m,*

$$\boxed{\omega(m(m+1)) \leq \iota(m(m+1))^{1/2+\varepsilon}.}$$

Proposition 2.4 (Equivalent maximal formulation)**.** *Conjecture 2.1 is equivalent to*

$$\boxed{W(r) = o(r).}$$

Likewise Conjecture 2.3 is equivalent to $W(r) \leq r^{1/2+o(1)}$.

Proof. For $m \in \mathcal{S}_r$, $\omega(m(m+1)) \leq W(r)$, and setting $r = r(m)$ gives the easy implication from $W(r)/r \to 0$ to $D(m) \to 0$. For the converse, fix $\varepsilon > 0$; the dilution hypothesis gives an M such that $s(m) < \varepsilon r(m)$ for $m > M$. When $m \leq M$, $s(m)$ is bounded independently of r. Thus for all $m \in \mathcal{S}_r$,

$$s(m) \leq \max\{\max_{m \leq M} s(m), \varepsilon r\},$$

and $W(r)/r \leq \varepsilon + o(1)$. For the power bound, the finitely many $m \leq M_\varepsilon$ are again harmless, while $r(m) \leq r$. □

The hierarchy is

$$\text{square-root dilution} \Longrightarrow \text{power-saving dilution} \Longrightarrow \text{dilution}$$
$$\Longrightarrow \text{eventual non-PC} \Longrightarrow \text{A141399 finite.}$$

The final implication needs care: $D(m) \to 0$ ensures that $D(m) = 1$ occurs only finitely often, but the converse is not known. In particular, disproving the square-root law would not by itself disprove A141399 termination.

3 What is provable unconditionally?

3.1 High-support pairs must be logarithmically smooth

Lemma 3.1 (Critical smoothness obstruction)**.** *Fix $0 < \delta \le 1$. Every sufficiently large m with $D(m) \ge \delta$ satisfies*

$$\boxed{P^+(m(m+1)) \le \left(\frac{2}{\delta} + o(1)\right) \log(m+1).}$$

The $o(1)$ is interpreted along sequences with $r(m) \to \infty$; for bounded $r(m)$ the bound is automatic for large m.

Proof. Write $r = r(m)$ and $s = s(m) \ge \delta r$. The s distinct primes of $m(m+1)$ have product at least P_s. Thus

$$2\log(m+1) > \log(m(m+1)) \ge \log P_s = \vartheta(p_s).$$

The prime number theorem yields, for fixed $\delta > 0$ and $r \to \infty$,

$$\vartheta(p_s) \ge \vartheta(p_{\lceil \delta r \rceil}) = (\delta + o(1)) p_r.$$

Rearranging proves the result. If r stays bounded, p_r is bounded and the displayed inequality holds trivially for sufficiently large m. □

This lemma is a concrete, unconditional consequence of the support fraction condition: a hypothetical infinite family retaining a fixed positive fraction of prime indices would force infinitely many consecutive pairs whose largest prime factor is at most a constant times the logarithm of their size. That is the xyz-critical scale, not a generic smoothness condition.

3.2 An unconditional subpolynomial exceptional set

Define

$$E_\delta(X) = \#\{1 \le m \le X : D(m) \ge \delta\}.$$

Theorem 3.2 (Sparsity of non-dilute support)**.** *For every fixed $\delta > 0$,*

$$\boxed{E_\delta(X) = X^{o(1)} \qquad (X \to \infty).}$$

More quantitatively, with a constant depending on δ,

$$E_\delta(X) \le \exp\left(O_\delta\left(\frac{\log X \, \log\log\log X}{\log\log X}\right)\right)$$

for sufficiently large X.

Proof. By Lemma 3.1, there is $C_\delta > 0$ such that every sufficiently large m counted by $E_\delta(X)$ is $C_\delta \log X$-smooth. Hence, for fixed δ,

$$E_\delta(X) \le O_\delta(1) + \Psi(X, C_\delta \log X),$$

where $\Psi(X, y)$ counts positive y-smooth integers at most X. For X large put $y = C_\delta \log X$ and $\sigma = 1/\log\log X > 0$. Rankin's elementary inequality gives

$$\Psi(X, y) \le X^\sigma \prod_{p \le y} (1 - p^{-\sigma})^{-1}.$$

Since $p \ge 2$ and $1 - 2^{-\sigma} \asymp \sigma$, the logarithm of the Euler product is at most

$$\pi(y) \log \frac{1}{1 - 2^{-\sigma}} = O_\delta\left(\frac{\log X}{\log\log X} \log\log\log X\right).$$

Moreover, $\sigma \log X = \log X/\log\log X$. The asserted bound follows on exponentiating. □

Remark 3.3 (What the theorem does *not* say). For any $\delta > 0$ the exceptional set has natural density zero and, in fact, grows more slowly than X^ε for every fixed $\varepsilon > 0$. Yet $X^{o(1)}$ may still tend to infinity. The theorem does *not* prove eventual absence of exceptions, dilution for every m, or the finiteness of A141399. The existential uniform problem is precisely where the existing methods lose control.

3.3 An unconditional lower bound on the smooth-pair maximum

Proposition 3.4 (Unbounded support capacity)**.** *As $r \to \infty$,*

$$\boxed{W(r) \ge (1 + o(1)) \frac{\log r}{\log\log r}.}$$

Proof. Let k be maximal with $P_k + 1 \le p_r$. The pair $(m, m+1) = (P_k, P_k + 1)$ is p_r-smooth, since every prime factor of either member is at most $P_k + 1 \le p_r$. Its members are coprime; therefore the product has at least $k + 1$ distinct prime factors. Consequently $W(r) \ge k + 1$. From $\log P_k = \vartheta(p_k) \sim k \log k$ and $p_r \sim r \log r$, maximality implies $k = (1 + o(1)) \log r/\log\log r$. □

This lower bound leaves an enormous interval between the known unconditional growth and the proposed square-root upper law.

4 Finite enumeration: direct evidence and its limits

We performed an exhaustive enumeration of every consecutive p_r-smooth pair for $1 \le r \le 20$, using the maximum starting values U_r tabulated by OEIS A002072 [22]. The counts were independently cross-checked against OEIS A002071 [23]. Enumerating all p_{20}-smooth positive integers at most $U_{20} = 119089041053696$, checking whether the successor is p_{20}-smooth, and recording the distinct-prime support gives $4,839$ consecutive pairs. The source code supplied with this note implements this exact integer enumeration; its output is supplied as a separate CSV file. The enumeration uses the published endpoints: it does not independently reprove the Størmer certificates determining those endpoints.

r	p_r	$\#\mathcal{S}_r$	$W(r)$	$W(r)/r$	example m attaining $W(r)$
1	2	1	1	1.000	1
2	3	4	2	1.000	3
3	5	10	3	1.000	5
4	7	23	4	1.000	35
5	11	40	5	1.000	539
6	13	68	6	1.000	1715
7	17	108	7	1.000	12375
8	19	167	8	1.000	633555
9	23	241	8	0.889	52325
10	29	345	9	0.900	2697695
11	31	482	10	0.909	3897165
12	37	653	10	0.833	6770555
13	41	869	11	0.846	30944913
14	43	1153	12	0.857	658831250
15	47	1502	13	0.867	1777331555
16	53	1930	13	0.813	1777331555
17	59	2454	14	0.824	1284729638049
18	61	3106	14	0.778	1284729638049
19	67	3896	15	0.789	5163068910
20	71	4839	15	0.750	5163068910

Table 1: Exhaustive smooth-neighbor support maxima for $r \leq 20$. The final column provides one witness, not necessarily the largest witness, of the maximum $W(r)$. The pair counts reproduce A002071; the cutoffs U_r are from A002072.

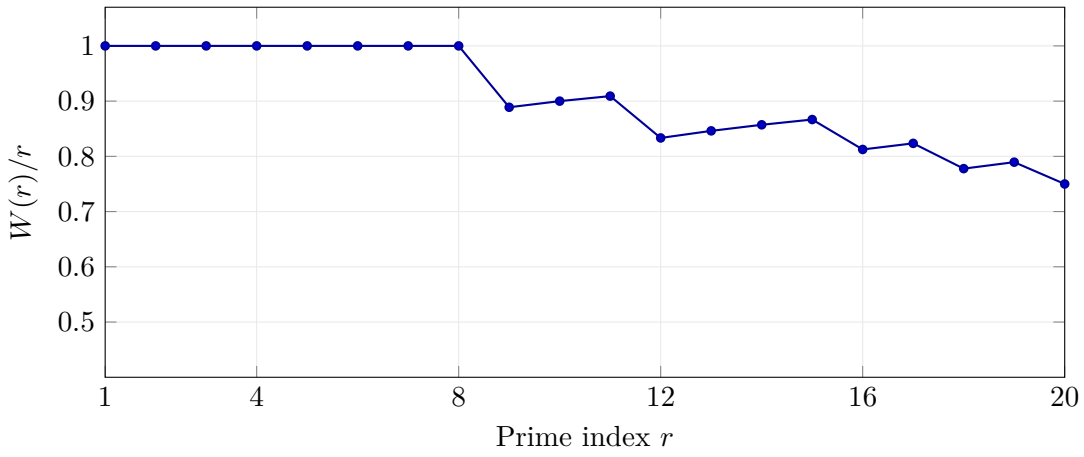


Figure 1: Finite maximal support fraction. The range is too small to identify an asymptotic exponent; conspicuous upward jumps remain.

These calculations support the observation that *full* support stops appearing after $r = 8$ over the tested range, but do not yet support a quantitative square-root exponent. Indeed, $W(20)/20 = 0.75$, much greater than $20^{-1/2} \approx 0.224$; the conjecture allows non-asymptotic constants and a remote threshold, so this is not a contradiction, but it is a warning against overinterpreting the data.

An individual pair may display far more dramatic dilution than the worst-case maximum. Three exactly factored examples are

$$\begin{aligned} m = 633555: &\quad (s, r) = (8, 8), \quad D = 1, \\ m = 10333245: &\quad (s, r) = (10, 548), \quad D \approx 0.01825, \\ m = 715041747420: &\quad (s, r) = (15, 1243236), \quad D \approx 0.00001207. \end{aligned}$$

For the second pair, the new prime factors are 2789 and 3947; for the third, 19540397 is the largest prime factor. These examples illustrate the mechanism but are *not representative samples* and cannot establish an eventual bound for all m.

5 Literature review and a classical conjectural source of the square-root law

5.1 Erdős Problem 368: the decisive prior-art connection

Put $F(m) = P^+(m(m+1))$. Erdős's long-standing Problem 368 asks how small $F(m)$ can be and, in particular, records the conjectural uniform lower bound $F(m) \gg (\log m)^2$; it also records Erdős's complementary expectation that $F(m) < (\log m)^{2+\varepsilon}$ infinitely often, for each $\varepsilon > 0$ [1, 2]. The lower-bound statement is not known unconditionally.

Theorem 5.1 (Conditional on the conjectured Erdős lower bound)**.** *If there is a constant $c > 0$ such that $F(m) \geq c(\log m)^2$ for all sufficiently large m, then*

$$\omega(m(m+1)) \leq \iota(m(m+1))^{1/2+o(1)}, \qquad W(r) \leq r^{1/2+o(1)}.$$

In particular, the prime-support dilution conjecture and finiteness of A141399 follow.

Proof. Let $y = F(m) = p_r$. The assumed bound gives $\log m \ll \sqrt{y}$. Since $m(m+1) < (m+1)^2$,

$$s = \omega(m(m+1)) \leq \frac{\log(m(m+1))}{\log 2} \ll \sqrt{y}.$$

As $r \to \infty$, $y = p_r = r^{1+o(1)}$ by the prime number theorem, proving $s \leq r^{1/2+o(1)}$. For the maximum $W(r)$, apply the same inequality to each $m \in \mathcal{S}_r$; small m contribute bounded s, while in the remaining range $\iota(m(m+1)) \leq r$. The conclusion follows. □

Proposition 5.2 (General power threshold)**.** *If for some fixed $\alpha > 1$ one has $F(m) \geq (\log m)^{\alpha-o(1)}$ as $m \to \infty$, then*

$$\omega(m(m+1)) \leq \iota(m(m+1))^{1/\alpha+o(1)}; \qquad W(r) \leq r^{1/\alpha+o(1)}.$$

Consequently any such exponent $\alpha > 1$ implies power-saving dilution and eventual absence of prime-complete products.

Proof. Use $\log m \le F(m)^{1/\alpha+o(1)}$, the elementary bound $\omega(m(m+1)) \ll \log(m+1)$, and $F(m) = p_{r(m)} = r(m)^{1+o(1)}$. The finite-small-$m$ argument from Theorem 5.1 gives the uniform maximum statement. □

This positions our square-root conjecture as a proposed consequence of a classical, more informative conjecture about the actual size of the greatest prime factor. The two conjectures are *not* asserted to be equivalent: small prime support can occur for reasons other than an unusually large greatest prime factor.

5.2 The uniform smooth-neighbor ceiling is an equivalent extremal problem

The sufficient condition proposed in §2, $\log U_r = o(p_r)$, has an exact reformulation.

Proposition 5.3 (Uniform-height/large-factor equivalence)**.** *The following two assertions are equivalent:*

$$\log U_r = o(p_r) \quad (r \to \infty),$$
$$\frac{F(m)}{\log m} \longrightarrow \infty \quad (m \to \infty).$$

Proof. For a given m set $r = \pi(F(m))$. Then $m \le U_r$, and $r \to \infty$ with m by fixed-S Størmer finiteness. Hence $\log m/F(m) \le \log U_r/p_r \to 0$. Conversely, $U_r \to \infty$ as $r \to \infty$ (for instance, $U_r \ge p_r - 1$). Put $m = U_r$. Then $F(m) \le p_r$, so $\log U_r/p_r \le \log U_r/F(U_r) \to 0$. □

This equivalence makes a concrete weaker target visible: $F(m)/\log m \to \infty$ already implies full dilution, though it does not imply the proposed $1/2$ exponent. A fixed power saving $F(m) \ge (\log m)^{1+\eta}$ is stronger and yields power-saving dilution. Erdős's quadratic-logarithm conjecture is stronger still.

5.3 What is established without abc or xyz?

The Erdős Problems survey records Pólya's theorem $F(m) \to \infty$, Mahler's bound $F(m) \gg \log\log m$, and the later improvement of Pasten [1, 3]. Specifically, Corollary 1.5 of Pasten's 2024 paper gives a lower bound for the greatest prime factor of $xy(x+y)$, for coprime positive integers $x < y$. Putting $(x, y) = (1, m)$ yields

$$F(m) \gg \frac{(\log\log m)^2}{\log\log\log m}.$$

This is still very far from $F(m)/\log m \to \infty$. Pasten's argument combines linear forms in logarithms with modular and Shimura-curve methods; it is a relevant source for our elliptic-geometry branch, but it does not furnish the desired uniform conclusion. His September 2026 follow-up improves a bound for $n^2 + 1$, *not* directly the bound for $m(m+1)$ [4]; these two polynomial problems must not be conflated.

5.4 Smooth blocks and distribution results: what they do and do not imply

Balog and Wooley proved that for any $\varepsilon > 0$ there exist infinitely many increasingly long blocks of consecutive integers of size roughly m whose prime factors do not exceed m^ε [5]. This does not contradict our conjecture: the function m^ε is much larger than any fixed power of $\log m$.

Work of Teräväinen on logarithmic averages, and later work of Tao–Teräväinen and others, investigates joint distributions of $P^+(m)$ and $P^+(m+1)$ in regimes where the smoothness bound is a fixed positive power of m [8, 9, 10]. These results substantially inform independence heuristics, but their asymptotic ranges do not authorize an independence assumption for $y = (\log m)^C$.

5.5 OpenAI Result 012: joint Dickman law for consecutive integers

An OpenAI preprint dated September 24, 2026, entitled *The joint Dickman law for consecutive integers*, appears as *Result 012: Independent largest prime factors of consecutive integers* in the October 2026 mathematics release [12, 13]. The manuscript claims the following ordinary-natural-density limit. For every fixed pair $0 < a, b < 1$,

$$\lim_{X\to\infty} \frac{1}{X} \#\Big\{2 \le m \le X : P^+(m) \le m^a, \quad P^+(m+1) \le m^b\Big\} = \rho(1/a)\rho(1/b), \tag{1}$$

where ρ is the Dickman function. It also claims that each of $P^+(m) < P^+(m+1)$ and $P^+(m) > P^+(m+1)$ holds for a set of natural density $1/2$. These assertions address the Erdős–Pomerance joint-distribution problem (catalogued as Erdős Problem 928) and the ordering problem (Problem 371), respectively [12, 15].

Status and formal verification. The repository advertises a Lean formalization covering the joint ordinary-density law and the two ordering statements, rather than all 84 pages of the manuscript. The visible comparator challenge `JointDickman.lean` is a statement template: its theorem bodies contain `sorry`, which must *not* be mistaken for a completed proof. A separate solution module is reported, and an independent October 2026 statement-and-build audit reports successful type checking without `sorry` and with the expected foundational axioms [14, 15]. We have checked the public statement, scope descriptions, and this audit, but have *not* independently replayed its Lean build or refereed the manuscript. We therefore describe the theorem here as a released, formally reported result, not as an independently audited theorem of this note.

Relevance and the quantifier obstruction. Taking $a = b = \alpha$ in (1) predicts, for every *fixed* $0 < \alpha < 1$, a positive proportion $\rho(1/\alpha)^2$ of pairs for which both integers are m^α-smooth. This is an important confirmation of smooth-neighbor correlations at fixed power scales. But our extremal problem needs a bound on *every* sufficiently large m, or on the last consecutive p_r-smooth pair for growing r. The crucial necessary scale in the sparsity argument is $P^+(m(m+1)) \ll \log m$; even a $(\log m)^C$ threshold is far smaller than m^α for every fixed $\alpha > 0$.

It is invalid to replace the fixed exponent α in (1) by $\alpha = \alpha(X) \to 0$ without a new uniform error estimate; moreover, a density-zero exceptional set could still be infinite. Thus Result 012 neither proves Erdős Problem 368 nor improves the $X^{o(1)}$ worst-case exception bound of Theorem 3.2. What it does provide is a powerful, particularly pertinent context for the Dickman product heuristic, and a promising body of proof techniques whose *uniformity* would need to be quantified afresh before use in Track 2.

Specific next test. Examine the release's decorrelation argument for explicit dependence on a, b and the size scale X; determine whether any uniformity survives when a, b shrink with X. A usable estimate at $P^+(m), P^+(m+1) \le (\log m)^C$ would be a genuinely new ingredient, but ordinary density alone would still not exclude an infinite sequence of exceptional smooth pairs. For eventual

termination one needs a uniform upper bound on their sizes or a support-sensitive obstruction that rules them out.

5.6 Classical S-unit theory and the obstacle of growing prime sets

Evertse and Győry's treatment of the equation $u + v = 1$ in a fixed finitely generated unit group gives effective finiteness and practical algorithms [6]; von Känel and Matschke use modularity and height/sieve techniques to solve extensive fixed-S families [7]. These results are directly relevant to Størmer enumeration and to numerical certification. Their existence does not supply an asymptotically uniform bound when S grows through the first r primes.

Lagarias and Soundararajan's xyz framework is treated in the next section [16, 11]. In our special case $m + 1 = m + 1$, the distinction between the general xyz exponent $3/2$ and Erdős's predicted exponent 2 for consecutive products should be made explicit.

5.7 Originality and proof-status assessment

We have not found the normalized support statistic $\omega(m(m+1))/\pi(P^+(m(m+1)))$ or the extremal $W(r)$ under these exact names in the surveyed literature. That is *not* a proof of novelty. The sparsity bound of Theorem 3.2 is an elementary consequence of standard prime-number and smooth-number estimates, and should not be presented as an independently established breakthrough without further checking. The strongest research contribution currently available is a clear bridge from prime-support statistics to Erdős Problem 368, with certified finite maxima $W(r)$ and a precise record of conditional dependencies. The addition of Result 012 is a literature connection, not a proof of the uniform conjecture. Its relation to Problem 368 is conceptual and methodological, not an implication of the theorem as stated.

6 A rigorous conditional bound from strong xyz

For a primitive triple $A + B = C$, write

$$H = \max(|A|, |B|, |C|), \qquad S = P^+(ABC), \qquad \kappa(A, B, C) = \frac{\log S}{\log \log H}.$$

The strong Lagarias–Soundararajan xyz conjecture states that the liminf of κ over primitive triples of increasing height is $3/2$ [16]. In particular, for any fixed $\varepsilon > 0$, all sufficiently large primitive triples satisfy $S > (\log H)^{3/2-\varepsilon}$.

Theorem 6.1 (Conditional on strong xyz)**.** *Assume strong xyz. Then*

$$\boxed{W(r) \le r^{2/3+o(1)}} \qquad (r \to \infty).$$

Consequently, for every $\varepsilon > 0$,

$$\omega(m(m+1)) \le \iota(m(m+1))^{2/3+\varepsilon}$$

for all sufficiently large m. In particular, the dilution conjecture follows conditionally.

Proof. Use the primitive triple $(m, 1, m+1)$, for which $H = m+1$ and $S = P^+(m(m+1))$. Strong xyz gives, for arbitrary small fixed $\varepsilon > 0$ and all sufficiently large m,

$$\log(m+1) < S^{1/(3/2-\varepsilon)}.$$

For $m \in \mathcal{S}_r$, $S \le p_r$. Since

$$\omega(m(m+1)) \le \frac{\log(m(m+1))}{\log 2} < \frac{2\log(m+1)}{\log 2},$$

we obtain $\omega(m(m+1)) \ll_\varepsilon p_r^{1/(3/2-\varepsilon)}$. Small m contribute only a bounded number of prime factors. Now $p_r \sim r\log r$, and $1/(3/2-\varepsilon) \to 2/3$ as $\varepsilon \downarrow 0$. Taking the maximum over $m \in \mathcal{S}_r$ proves the claim. □

Remark 6.2 (How much xyz is really needed?). An eventual lower bound $P^+(m(m+1)) \ge (\log m)^{1+\eta}$, with any fixed $\eta > 0$, already gives $W(r) \le r^{1/(1+\eta)+o(1)}$. Thus an xyz-type exponent *strictly greater than 1* is sufficient for power-saving dilution. The full conjectured exponent $3/2$ is not necessary. The usual direct application of *abc* has critical exponent 1 and does not yield this power saving.

7 Why a square-root exponent is plausible but unproved

Let $y = p_r$, and suppose one models the chance that an integer of size x is y-smooth using the Dickman function $\rho(u)$, where $u = \log x/\log y$. If the smoothness of adjacent integers behaved approximately independently, a heuristic expected count of smooth pairs through x would have scale

$$x\,\rho(u)^2, \qquad \log\rho(u) \sim -u\log u \quad (u\to\infty) \quad [19, 20].$$

Set $\log x = y^\alpha$. Then

$$\log\bigl(x\rho(u)^2\bigr) \sim (1-2\alpha)y^\alpha,$$

which changes sign at $\alpha = 1/2$. This is consistent with Erdőss independently stated quadratic-logarithm conjecture in Section 5, and motivates the *heuristic* ceiling

$$\log U_r \le p_r^{1/2+o(1)},$$

and hence $W(r) \le r^{1/2+o(1)}$.

There are two serious caveats. The approximation $x\rho(u)^2$ need not be valid uniformly in the delicate extreme range used here; in particular, the events of consecutive smoothness are arithmetically correlated. Even a correct estimate for a typical interval would not exclude extremely rare exceptional pairs. The square-root law is therefore an explicit conjectural target, *not* a consequence of standard Dickman asymptotics.

8 Possible proof strategies independent of xyz

8.1 Uniform height bounds for smooth neighbors

A particularly transparent sufficient theorem would be

$$\boxed{\log U_r = o(p_r).} \tag{2}$$

Indeed, if $s \geq \delta r$ for infinitely many p_r-smooth consecutive pairs, the proof of Lemma 3.1 would force $\log U_r \geq (\delta/2 + o(1))p_r$, contradicting (2). Thus (2) implies $W(r) = o(r)$.

A still stronger, potentially more approachable *quantitative* form is

$$\log U_r \ll p_r^{1-\eta} \quad \text{for some } \eta > 0.$$

It implies power-saving dilution by the same argument. But presently known effective S-unit bounds and Pell-equation methods do not provide this uniformity as the number of allowed primes grows. For comparison, the weaker Track 1 sufficient inequality $U_r(U_r + 1) < P_r$ corresponds roughly to $\log U_r < \frac{1}{2}p_r$; it can establish PC termination without forcing full dilution.

8.2 Support-sensitive S-unit geometry

Instead of bounding the entire set of consecutive p_r-smooth pairs, one could study those with support at least δr. Write $m = \prod_{p \in A} p^{a_p}$, $m + 1 = \prod_{p \in B} p^{b_p}$ with disjoint A, B, and assume $|A| + |B| \geq \delta r$. A hypothetical uniform theorem excluding such high-support S-unit solutions beyond an order $R(\delta)$ would directly imply dilution. One could seek a bound involving the number and distribution of the distinct primes, not merely their maximum. The key missing step is to make that obstruction uniform over all disjoint partitions and exponent vectors.

8.3 CRT geometry, combined with multiplicative rigidity

For fixed r, divisibility by every prime through p_r gives 2^r CRT classes modulo P_r. More general support-rich solutions need only s of the first r primes; for each chosen support set, congruence conditions select locations of the multiples. CRT alone cannot solve the problem: prescribed finite divisibility patterns recur periodically without limit. To exploit this geometry one must prove that the surviving positions eventually acquire unwanted larger prime factors, or that the required prime-power products cannot differ by one. A rigorous multi-scale sieve plus an S-unit rigidity lemma would constitute genuine progress.

8.4 A quantitative exceptional-set bridge

Theorem 3.2 is a useful unconditional starting point, but the bound $E_\delta(X) = X^{o(1)}$ leaves room for infinitely many exceptional m. An ambitious next step is to study the sets $E_\delta(X)$ not only by their size but by their Diophantine structure. Any theorem making the bound *absolute*, uniformly in X, would establish dilution. This is a large step beyond ordinary smooth-number counts, so a useful intermediate objective would be an effective classification of exceptional families or a structural obstruction applying to all sufficiently large members.

8.5 What computational experiments can falsify

Extend the $W(r)$ census using the existing Størmer–Lehmer/Pell certificates (for example, through the already computed smoothness orders beyond 20), recording not just $W(r)$ but every maximizing witness, its support partition, and the prime indices actually missing. Plot $W(r)/r$ and $\log W(r)/\log r$, but do not estimate an asymptotic exponent from a range on which the ratio is nonmonotone. A counterexample to any proposed finite-range inequality is useful even when it does not contradict the asymptotic conjecture. Existing Track 1 certificates through $r = 37$ can

cross-check the exclusion of full support; they do not automatically supply the exact values of $W(r)$.

9 Summary of proved, conditional, and conjectured claims

Statement	Status
$D(m) = 1$ iff $m(m+1)$ is prime-complete	Elementary theorem
$D(m) \to 0$ iff $W(r) = o(r)$	Elementary theorem
$E_\delta(X) = X^{o(1)}$ for every fixed $\delta > 0$	Unconditional theorem
$W(r) \gg \log r / \log\log r$	Unconditional theorem
Values of $W(r)$ for $1 \le r \le 20$	Exhaustive finite computation using published U_r
$W(r) \le r^{2/3+o(1)}$	Conditional on strong xyz
$W(r) = o(r)$	Conjecture; open
$W(r) \le r^{1/2+o(1)}$	Stronger heuristic conjecture; open
A141399 is finite	Open unconditionally

10 Research provenance and the limits of an originality claim

The prime-support dilution conjecture in this note arose when a human researcher, who had been studying A141399 and consecutive smooth pairs, asked an LLM to propose a new conjecture about prime-complete numbers. The model proposed the normalized support fraction $D(m)$, the limit $D(m) \to 0$, and a speculative square-root upper exponent for the extremal function $W(r)$. Subsequent collaboration formulated a reproducible finite test, proved the elementary sparse-exception estimate, and searched the literature. The search uncovered the strong connection to Erdős Problem 368 and, later, Result 012.

This chronology is evidence that a language model can generate a mathematically specific, testable conjectural direction beyond simply answering the problem initially posed to it. It does *not* by itself establish historical originality, truth of the conjecture, or that the model is capable of supplying a valid proof. The square-root law is already implied by Erdős's older conjectured bound, and the $X^{o(1)}$ sparsity derivation uses standard techniques. Its value as a research example lies in the iterative sequence of generation, mathematical scrutiny, computational falsifiability, and discovery of connections to prior work.

Verification and understanding are separate stages. A convincing argument needs reconstruction and proof checking; a purportedly original problem needs a prior-art search. For the present manuscript, no unconditional proof of uniform dilution, eventual A141399 termination, or the square-root bound is claimed.

11 Conclusion

The central conjecture proposes that consecutive products eventually occupy a vanishing fraction of the prime indices through their greatest prime factor. An unconditional theorem shows that every fixed positive fraction is already extraordinarily rare: only $X^{o(1)}$ starting values up to X can attain it. The remaining difficulty is to turn exceptional *sparsity* into uniform *impossibility*. Strong

xyz would achieve much more, giving an exponent-2/3 bound; the square-root exponent suggested by a random smooth-neighbor model is stronger still and remains speculative. The computed data through $r = 20$ justify continued investigation, but they do not yet establish the proposed decay rate. The literature review identifies the conjectural Erdős quadratic-logarithm bound as a sufficient route to our strongest variant; weaker bounds $F(m)/\log m \to \infty$ would already prove dilution. A proof independent of xyz would likely need a new support-sensitive uniform result about consecutive S-units, or a much sharper height bound for consecutive smooth pairs.

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